

VECTOR BUNDLES ON THE FARGUES-FONTAINE CURVE

1. THE MAIN THEOREM

In this talk, I will be focusing on the classification of vector bundles on the Fargues-Fontaine curve following the exposition in Fargues-Scholze [1].

First, we will need to construct some of the relevant vector bundles. Throughout, let E be a finite extension of $\mathbf{Q}_p$ with residue field $\mathbf{F}_q$, ring of integers $\mathcal{O}_E$ and a choice of uniformizer π . We will also put C as an algebraically closed perfectoid field over $\mathbf{F}_q$, and denote $X_{C,E}$ as X_C as E is implicit.

As a means for constructing vector bundles, we will use the category of isocrystals.

DEFINITION 1.1. Let $E/\mathbf{Q}_p$, and put $\check{E} = W_{\mathcal{O}_E}(\overline{\mathbf{F}}_q)[1/\pi]$ for the maximal unramified extension.

The category $\mathbf{Isoc}_E$ is the E -linear $\otimes$ -category with objects (V, φ) where $V \in \mathbf{Vect}_{\check{E}}$ and $\varphi : V \simeq V$ is a $\varphi_{\check{E}}$ -semilinear isomorphism.

For $\lambda = m/n \in \mathbf{Q}$ for m, n coprime and $n > 0$, we set V_λ to be the isocrystal with vector space $\check{E}^n$ and semilinear automorphism

$$\begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ \pi^m & & & 0 \end{pmatrix} \varphi_{\check{E}}.$$

There is a functor

$$\mathbf{Isoc}_E \rightarrow \mathbf{Vect}(X_C)$$

arising from the observation that

$$Y_{C,E} \rightarrow \mathrm{Spa} \check{E}$$

and the structure morphism is equivariant for φ_C acting on $Y_{C,E}$ and $\varphi_{\check{E}}$ acting on $\mathrm{Spa} \check{E}$. Indeed, this induces a pullback functor

$$\mathbf{Isoc}_E = \mathbf{Vect}^{\varphi_{\check{E}}}(\mathrm{Spa} \check{E}) \rightarrow \mathbf{Vect}^{\varphi_C}(Y_{C,E}).$$

But then by descent the latter is just $\mathbf{Vect}(X_C)$.

DEFINITION 1.2. We set $\mathcal{O}(\lambda)$ to be the image of $V_{-\lambda}$ under this map, so that $\mathcal{O}(1)$ is ample.

We are now ready to state the main theorem.

THEOREM 1.3 (Fargues-Fontaine classification, [2]). There is a decomposition

$$\mathcal{E} \simeq \bigoplus_{\lambda \in \mathbf{Q}} \mathcal{O}(\lambda)^{n_\lambda}$$

for any vector bundle $\mathcal{E} \in \mathbf{Vect}(X_C)$.

Recalling the functor

$$\mathbf{Isoc}_E \rightarrow \mathbf{Vect}(X_C)$$

sends $V_{-\lambda} \mapsto \mathcal{O}(\lambda)$, the Dieudonné-Manin classification says any isocrystal is isomorphic to a finite direct sum

$$\bigoplus_{\lambda \in \mathbf{Q}} V_{-\lambda}^{\oplus n_\lambda},$$

and hence, together with the classification theorem above, this functor induces a bijection on isomorphism classes.

REMARK 1.4. This generalizes to G -bundles and G -isocrystals for a reductive group G . We can interpret a G -torsor on X_C as an exact $\otimes$ -functor

$$\mathbf{Rep}_{\mathbf{Q}_p}(G) \rightarrow \mathbf{Vect}(X_C).$$

Then understanding $\mathbf{Vect}(X_C)$ sufficiently well, i.e. the previous decomposition, produces a functor

$$\mathbf{Rep}_{\mathbf{Q}_p}(G) \rightarrow \mathbf{Q} - \mathbf{FilVB}(X_C)^{\mathbf{HN}},$$

the category of $\mathbf{Q}$ -filtered vector bundles on X_C such that the $\lambda \in \mathbf{Q}$ component $\mathcal{E}^\lambda$ is semistable of slope λ . It's easy to check this is an exact $\otimes$ -functor: exactness follows from $\mathbf{Rep}_{\mathbf{Q}_p}(G)$ being semisimple, and we can use the previous classification of vector bundles to check it is a $\otimes$ -functor.

This allows us to produce an associated graded exact $\otimes$ -functor

$$\mathbf{Rep}_{\mathbf{Q}_p}(G) \rightarrow \mathbf{Isoc}_{\mathbf{Q}_p},$$

which is precisely the data of a G -isocrystal in $B(G)$. To show this classifies the isomorphism classes of vector bundles we just need to split the previous filtration, which is done by computing $H^1(X_C, \mathcal{O}(\lambda)) = 0$ for $\lambda > 0$ so there are no extensions.

2. AMPLENESS OF $\mathcal{O}(1)$

In the last section, we defined $\mathcal{O}(1)$ to be the image of the isocrystal V_{-1} under the functor

$$\mathbf{Isoc}_E \rightarrow \mathbf{Vect}(X_C).$$

It will be important for the argument to verify that $\mathcal{O}(1)$ is ample, or that $\mathcal{E}(n)$ is globally generated and $H^1(\mathcal{E}(n)) = 0$ for $n \gg 0$.

The reason we care about this is that it will give an injection

$$\mathcal{O}_{X_C}(-d) \rightarrow \mathcal{E}$$

for an arbitrary vector bundle. Indeed, a sufficiently large twist of $\mathcal{E}$ will then be globally generated and in particular admit a section, so upon untwisting we get the desired map.

THEOREM 2.1 (Kedlaya-Liu, [6, Section 8.7]). Let $S/\mathbf{F}_q$ be an affinoid perfectoid space $\mathrm{Spa}(\mathbf{R}, \mathbf{R}^+)$, and let $\mathcal{E}$ be a vector bundle on $X_{S,E}$. Then there is some n_0 such that for all $n \geq n_0$ the vector bundle $\mathcal{E}(n)$ is globally generated and $H^1(X_{S,E}, \mathcal{E}(n)) = 0$.

Sketch. The proof is quite complicated and technical, so we will only give the basic idea of how to approach the question. We'll focus on showing H^1 vanishes.

Noting that the Frobenius φ_S multiplies the radius by q , so we can present

$$X_S = Y_S / \varphi^{\mathbf{Z}} = Y_{S,[1,q]} / (Y_{S,[1,1]} \sim Y_{S,[q,q]}).$$

Here, $Y_{S,I}$ is the open affinoid annulus $\mathrm{rad}^{-1}(I)$ for the radius function

$$\mathrm{rad} : |Y_S| \rightarrow (0, \infty).$$

Explicitly,

$$Y_{S,[a,b]} = \{|\pi|^b \leq |[\varpi]| \leq |\pi^a|\} \subset Y_S.$$

An immediate consequence of this presentation is that upon building a Čech complex computing cohomology, one obtains

$$R\Gamma(X_S, \mathcal{E}) = [\mathcal{E}(Y_{S,[1,q]}) \rightarrow \mathcal{E}(Y_{S,[q,q]})]$$

via $\varphi_S - 1$. By vanishing for affinoids, with no work we see H^2 vanishes. To get H^1 to vanish, you need to check this map is surjective for a sufficiently large twist.

The Čech approach allows us to reduce this to a commutative algebra question: any $\mathcal{E}$ can be written a finite projective $B_{R,[1,q]}$ -module M with an isomorphism on its base changes

$$\varphi_M : M_{[q,q]} \simeq M_{[1,1]}$$

which is linear over φ .

Kedlaya-Liu show that one can reduce to the case where M is free, and in this case φ_M is given quite explicitly by

$$\varphi_M = A^{-1}\varphi$$

for $A \in \mathrm{GL}_m(B_{R,[1,1]})$. Under this description of a vector bundle, a twist by $\mathcal{O}(1)$ amounts to sending $A \mapsto A\pi$ (recall π is the uniformizer for E ; we use ϖ for perfectoid spaces). Once this setup is done, Kedlaya-Liu manually check global generation by producing explicit elements and verify $\varphi - A$ is surjective after an appropriate twist to manipulate the matrix entries.

More precisely, they show that for $1 < r \leq q$ rational there are m elements

$$v_1, \dots, v_m \in (B_{R,[1,q]}^m)^{\varphi=A} = H^0(X_S, \mathcal{E})$$

which form a basis of $B_{R,[r,q]}^m$. Applying this to enough strips proves global generation, and one proves this by showing $\varphi - A$ is surjective in an *effective* way, that is one can pick preimages for $\varphi - A : B_{R,[1,q]}^m \rightarrow B_{R,[1,1]}^m$ such that the preimage has a small spectral norm on $B_{R,[r,q]}^m$. Kedlaya-Liu provide a convergent procedure to produce these preimages, and then pick $v_i = [\varpi]^M e_i - v'_i$ as small perturbation of the standard basis to land in the $\varphi = A$ invariants. Here, v'_i is chosen so $(\varphi - A)(v'_i) = (\varphi - A)([\varpi]^M e_i)$ (thus landing in the $\varphi = A$ fixed points) but has a sufficiently small norm on $B_{R,[1,q]}^m$ so that these remain a basis. $\square$

3. THE HN FORMALISM

We will begin by recalling what the Harder-Narasimhan formalism is for a curve $X/\mathbf{C}$.

DEFINITION 3.1. Let $\mathcal{E}$ be a vector bundle on a smooth projective curve $X/\mathbf{C}$. We define the *slope* of $\mathcal{E}$ to be $\lambda = \deg(\mathcal{E})/\mathrm{rank}(\mathcal{E}) \in \mathbf{Q}$.

A vector bundle is *semistable* if any proper nonzero subbundle $\mathcal{E}'$ has $\lambda(\mathcal{E}') \leq \lambda(\mathcal{E})$.

THEOREM 3.2. Let $\mathcal{E}$ be a vector bundle on a smooth projective curve $X/\mathbf{C}$. Then there exists a unique filtration

$$0 = \mathcal{E}_0 \subset \dots \subset \mathcal{E}_r = \mathcal{E}$$

such that all subquotients $\mathcal{F}_i = \mathcal{E}_{i+1}/\mathcal{E}_i$ are semistable and slopes of $\mathcal{F}_i$ decrease as the index i increases.

As it turns out, an extremely similar formalism can be defined on the Fargues-Fontaine curve X_C . The non-obvious part of the definition of a slope is defining the degree, which requires us to determine the line bundles.

PROPOSITION 3.3. Let $S^\#$ be a characteristic zero untilt lying over E_∞ , the completion of the maximal abelian extension of E . Then there is an exact sequence of $\mathcal{O}_{X_S}$ -modules

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}(1) \longrightarrow \mathcal{O}_{S^\#} \longrightarrow 0.$$

Sketch. This is used several times, so I will explain how to write down the maps.

Providing a map $\mathcal{O} \rightarrow \mathcal{O}(1)$ amounts to taking the data of an untilt $S^\#$ and then providing a section $s \in H^0(X_{S,E}, \mathcal{O}_{X_{S,E}}(1))$. Using a slight modification of the Čech covering we used to show $\mathcal{O}(1)$ is ample, we can identify

$$H^0(X_{S,E}, \mathcal{O}_{X_{S,E}}(1)) = \mathcal{O}(Y_{[1,\infty]})^{\varphi=\pi}$$

where $Y_{[1,\infty]} = \{[|\varpi|] \leq |\pi| \neq 0\} \subset \mathrm{Spa} W_{\mathcal{O}_E}(S^+)$. Note that this is not contained in Y . Using the fact that Frobenius scales the radius function by q , we can further identify

$$\mathcal{O}(Y_{[1,\infty]})^{\varphi=\pi} = (B_{\mathrm{cris}}^+)^{\varphi=\pi}.$$

Now apply Scholze-Weinstein Theorem A: the Dieudonné functor on semiperfect rings is fully faithful [7]. We obtain

$$H^0(X_{S,E}, \mathcal{O}_{X_{S,E}}(1)) = \mathrm{Hom}_{\mathcal{O}_E}(E/\mathcal{O}_E, G(S^{\#+}/\pi))[1/\pi] = \tilde{G}(S^{\#+}/\pi) = \tilde{G}(S^{\#+})$$

where $G \simeq \mathrm{Spf} \mathcal{O}_E[[X]]$ is the Lubin-Tate formal group of E and $\tilde{G} = \varprojlim_{\times \pi} G \simeq \mathrm{Spf} \mathcal{O}_E[[\tilde{X}^{1/p^\infty}]]$ is the universal cover. In particular, the first identification we use the p -divisible group $E/\mathcal{O}_E$ and identify $G(S^{\#+}/\pi)$ with the points of the associated p -divisible group (by taking the p -adic Tate module for the formal group).

With this machinery in place, so long as our untilt lies over E_∞ we can produce a distinguished element of $\tilde{G}(C^{\#,+})$ via the map

$$V_\pi(G) \rightarrow \tilde{G}$$

where V_π is the rational π -adic Tate module. This arises by taking universal covers on $\bigcup_n G[\pi^n] \rightarrow G$. Given an untilt $C^\# / E_\infty$, we can produce an element of V_π which we use for the section.

The final map is just evaluation at $C^\#$. Exactness ends up being possible to reduce to $C^\#$ to the universal case of E_∞ where it can be checked directly. $\square$

PROPOSITION 3.4. Let x be a characteristic zero untilt. The scheme $X_C^{\mathrm{alg}} - [x]$ is affine, and the spectrum of a PID.

Now we can prove the following.

PROPOSITION 3.5. We have

$$\mathbf{Z} \simeq \text{Pic}(X_C)$$

via $n \mapsto \mathcal{O}(n)$.

Proof. First, by GAGA we may instead consider the algebraic curve. The corollary shows any vector bundle on X_C^{alg} is trivialized on $X_C^{\text{alg}} - [x]$, so any vector bundle is of the form $\mathcal{O}(n[x])$. Here we are appealing to the fact that the local ring at x is a DVR, so by Beauville-Laszlo gluing we have at the level of groupoids [3]

$$\text{Pic}(X_C^{\text{alg}}) \simeq \text{Pic}(X_C^{\text{alg}} - [x]) \times_{\text{Pic}(D_x^\circ)} \text{Pic}(D_x)$$

where $D_x = \widehat{\mathcal{O}_{X_C, x}}$ and D° punctures this. Knowing the local ring is a DVR, we get $\mathbf{Z}$. For example, if we had $\mathbf{C}_p[[t]]$ we look at $\mathbf{C}_p[[t]]$ lattices in $\mathbf{C}_p((t))$, which are classified by t^n . In general if R is a DVR with fraction field K these are going to be classified by $K^\times/R^\times$, or the value group, which is $\mathbf{Z}$.

As this point, we already know $\text{Pic}(X_C) \simeq \mathbf{Z}$, but the isomorphism is not canonical.

It suffices to show $\mathcal{O}([x]) \simeq \mathcal{O}(1)$ for any untilt $x = \text{Spa}(C^\sharp, C^{\sharp,+})$ to make the isomorphism canonical.

To see this look at the previous exact sequence

$$0 \longrightarrow \mathcal{O}_{X_C} \longrightarrow \mathcal{O}_{X_C}(1) \longrightarrow \mathcal{O}_{C^\sharp} \longrightarrow 0.$$

This means that the map $\mathcal{O}_{X_C} \rightarrow \mathcal{O}_{X_C}(1)$ factors through the twisted ideal sheaf $I_{[x]}(1)$, and by exactness it is an isomorphism as $I_{[x]}(1) = \ker(\mathcal{O}_{X_C}(1) \rightarrow \mathcal{O}_{C^\sharp})$.

Observe $I_{[x]}$ is just $\mathcal{O}(-[x])$. As we just argued that

$$I_{[x]}(1) \simeq \mathcal{O}_{X_C}$$

so in particular $\mathcal{O}(-[x]) \simeq \mathcal{O}_{X_C}(-1)$ by untwisting. Taking duals, the claim follows. $\square$

DEFINITION 3.6. Let $\mathcal{E}$ be a vector bundle on X_C . We define $\deg(\mathcal{E}) = \deg(\det \mathcal{E})$, where $\deg$ is the isomorphism $\text{Pic}(X_C) \rightarrow \mathbf{Z}$.

Then we set the slope λ to be the degree over the rank.

One can axiomatize a Harder-Narasimhan formalism and verify the axioms hold to deduce that it holds for X_C given this definition of a slope. This can be generalized, but the definition below is sufficient.

DEFINITION 3.7 (Abstract HN formalism). An abstract HN formalism consists of a quasi-abelian category $\mathcal{C}$ equipped with degree and rank functions $|\mathcal{C}| \rightarrow \mathbf{Z}$ and $\mathbf{Z}_{\geq 0}$ respectively which are additive on exact sequences.

REMARK 3.8. If $X = \operatorname{Spec} R$ is a scheme, generally $\mathbf{Vect}(X)$ is not quasi-abelian since we don't need to have kernels and cokernels (the third condition is that Ext is bifunctorial). But in the case that R is a Dedekind domain, like with a curve, this is true.

It's easy to check rank and degree are additive in short exact sequences. For rank this is clear, and for degree we just take determinant bundles: the determinant functor factors through $K^0(\mathbf{Vect}(X_C))$, so in particular for an exact sequence $0 \rightarrow \mathcal{E}_1 \rightarrow \mathcal{E}_2 \rightarrow \mathcal{E}_3 \rightarrow 0$ we obtain $\det(\mathcal{E}_2) \simeq \det(\mathcal{E}_1) \otimes \det(\mathcal{E}_3)$, and hence the degree is additive.

We then obtain the following corollary.

COROLLARY 3.9. The scheme X_C^{alg} has a Harder-Narasimhan formalism. That is, Theorem 3.2 holds verbatim with the definition of semistable being the same.

REMARK 3.10. The vector bundle $\mathcal{O}(\lambda)$ is always stable of slope λ , and $\mathcal{O}(\lambda)^n$ is always semistable of slope λ , or lies in $\mathbf{Vect}^\lambda(X_{C,E})$.

REMARK 3.11. If $\mathcal{E} \simeq \bigoplus_{\lambda \in \mathbf{Q}} \mathcal{O}(\lambda)^{n_\lambda}$, the slope of $\mathcal{E}$ is the rank-weighted average of the λ 's that appear:

$$\mu(\mathcal{E}) = \frac{\sum_{\lambda} n_{\lambda} \operatorname{rank}(\mathcal{O}(\lambda)) \lambda}{\sum_{\lambda} n_{\lambda} \operatorname{rank}(\mathcal{O}(\lambda))}.$$

4. REDUCTIONS FOR THE MAIN THEOREM

Having now established the Harder-Narasimhan formalism on $\mathbf{Vect}(X_C)$, we will be able to reduce the desired classification theorem to the case of semistable vector bundles and further to semistable slope 0 vector bundles. The triviality of semistable slope 0 vector bundles will be the most difficult part.

PROPOSITION 4.1. The cohomology group

$$H^1(X_S, \mathcal{O}_{X_S}(\lambda))$$

is trivial for $\lambda \in \mathbf{Q}_{>0}$. In particular, $\text{Ext}^1(\mathcal{O}(\lambda'), \mathcal{O}(\lambda)) = 0$ when $\lambda > \lambda'$.

Proof. This can be proven directly. First, we make a small reduction: if $\lambda = s/r$, replacing E with an unramified extension E'/E of degree r gives a covering $f : X_{S,E'} \rightarrow X_{S,E}$ of the curve where $f_*\mathcal{O}(s) = \mathcal{O}(s/r)$. Then it suffices to show vanishing for H^1 of $\mathcal{O}(n)$.

Recalling the module setup used to prove $H^1(X_C, \mathcal{E}(n)) = 0$ for $n \gg 0$, the corresponding object for $\mathcal{O}(n)$ is a free rank 1 module M over $B_{C,[1,q]}$ equipped with

$$\varphi_M = A^{-1}\varphi : M_{[q,q]} \rightarrow M_{[1,1]}.$$

Here A is an automorphism of $B_{R,[1,1]}$. Recall that using this presentation of X_S we get H^0 as the $\varphi = A$ invariants, since we need $M_{[q,q]}$ and $M_{[1,1]}$ to be identified. For higher cohomology we look at the derived invariants.

Since twisting corresponds to multiplication by π , we're looking at $A = \pi^n$. To get H^1 to vanish we'll need to show

$$\varphi - \pi^n : B_{C,[1,q]} \rightarrow B_{C,[1,1]}$$

is a surjection.

This can be done fairly directly, without the more involved methods Kedlaya-Liu used for surjectivity. Any element of $B_{C,[1,1]}$ has a decomposition into $B_{C,[0,1]}[1/\pi]$ and $[\varpi]B_{C,[1,\infty]}$. Here,

$$Y_{C,[0,1]} = \{|\pi| \leq |[\varpi]| \neq 0\}$$

and $[1, \infty]$ does the reverse; $[1, 1]$ asks for equality, which is why we have the decomposition.

Assume $f \in B_{C,[0,1]}$. Then

$$g = \varphi^{-1}(f) + \pi^n \varphi^{-2}(f) + \pi^{2n} \varphi^{-3}(f) + \dots$$

converges in $B_{[0,q]}$. Then g is an explicit preimage for f ; similarly this works for $B_{C,[0,1]}[1/\pi]$. For $[\varpi]B_{C,[1,\infty]}$ we use

$$g = -\pi^{-n}f - \pi^{-2n}\varphi(f) - \dots$$

which converges in $B_{C,[1,q]}$. Thus, we get explicit preimages.

To see the second claim, suppose we have an extension

$$0 \longrightarrow \mathcal{O}(\lambda) \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(\lambda') \longrightarrow 0.$$

Then $H^1(X_C, \mathcal{O}(\lambda) \otimes \mathcal{O}(\lambda')^\vee)$ parameterizes extensions. To see this is 0, it suffices to see $H^1(X_C, \mathcal{O}(\lambda - \lambda')) = 0$, which holds when $\lambda > \lambda'$ by the first claim. $\square$

PROPOSITION 4.2. We have

$$\mathrm{Ext}^1(\mathcal{O}_{X_C}(\lambda), \mathcal{O}_{X_C}(\lambda)) = 0$$

for any $\lambda \in \mathbf{Q}$.

Proof. As seen in the previous proposition, this amounts to $H^1(X_C, \mathcal{O})$ vanishing.

We can again appeal to the exact sequence

$$0 \longrightarrow \mathcal{O}_{X_S} \longrightarrow \mathcal{O}_{X_S}(1) \longrightarrow \mathcal{O}_{S^\#} \longrightarrow 0$$

for an untilt. In this case after taking cohomology we get

$$H^0(X_S, \mathcal{O}_{X_S}(1)) \xrightarrow{\log} S^\# \longrightarrow H^1(X_S, \mathcal{O}_{X_S})$$

where the first map is given by the logarithm map

$$\tilde{G}(S^{\#+}) \rightarrow G(S^{\#+}) \rightarrow S^\#$$

where G is the Lubin-Tate formal group and $\tilde{G}$ is the universal cover. Once we have identified $\tilde{G}(S^{\#+})$ with global sections of $\mathcal{O}(1)$ via Scholze-Weinstein theorem A, Lemma 3.5.1 of Scholze-Weinstein shows compatibility with the quasilogarithm [7, Lemma 3.5.1]. Unwinding definitions shows explicitly what the map to $S^\#$ is, and this map is pro-étale locally surjective with kernel $\underline{E}$. This shows $H^1(X_C, \mathcal{O})$ vanishes (we can identify it with a Banach-Colmez space so one may check on perfectoids). $\square$

COROLLARY 4.3. To deduce $\mathcal{E} \simeq \bigoplus_{\lambda \in \mathbf{Q}} \mathcal{O}(\lambda)^{n_\lambda}$ for any $\mathcal{E} \in \mathbf{Vect}(X_{C,E})$, it suffices to prove any semistable slope 0 vector bundle admits an injective map $\mathcal{O} \rightarrow \mathcal{E}$.

Proof. We argue by induction on the rank. Having computed $\mathrm{Pic}(X_C) \simeq \mathbf{Z}$ via $n \mapsto \mathcal{O}_{X_C}(n)$, we know the rank one case is done.

Next, suppose the theorem is proven for rank n and let $\mathcal{E}$ be of rank $n + 1$. If $\mathcal{E}$ is not semistable, then looking at the HN filtration

$$0 = \mathcal{E}_0 \subset \dots \subset \mathcal{E}_r = \mathcal{E}$$

we know $r - 1 \neq 0$. Thus, we look at $\mathcal{E}_{r-1}$, knowing that $\mathcal{E}/\mathcal{E}_{r-1}$ is a semistable vector bundle. That is, we obtain an extension

$$0 \longrightarrow \mathcal{E}_{r-1} \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}/\mathcal{E}_{r-1} \rightarrow 0,$$

where by induction on both sides the vector bundles are a direct sum of $\mathcal{O}(\lambda)$'s (and on the right, only the minimal slope λ). The first proposition then suffices to show $\mathcal{E}$ is a direct sum of $\mathcal{O}(\lambda)$'s.

Thus, we are reduced to the case where $\mathcal{E}$ is semistable of slope λ . By the second proposition, to deduce the claim amounts to producing an injective map

$$\mathcal{O}(\lambda) \rightarrow \mathcal{E}$$

since the category of semistable slope λ vector bundles is abelian (this is true in a general Harder-Narasimhan formalism); by applying the induction hypothesis, we see $\mathcal{E}$ is an extension of $\mathcal{O}(\lambda)$ and $\mathcal{O}(\lambda)^{\text{rank}(\mathcal{E})-1}$, which is necessarily trivial.

Finally, we reduce to the semistable slope 0 case. Let $\lambda = \frac{s}{r}$, and put E' as the unramified degree r extension of E . Consider the degree r covering

$$f : X_{C,E'} \rightarrow X_{C,E}.$$

Then $\mathcal{O}(\lambda) = f_*\mathcal{O}(s)$, so by adjunction we need a nonzero map

$$\mathcal{O}_{X_{C,E'}}(s) \rightarrow f^*\mathcal{E}.$$

Then by twisting we reduce to the slope 0 case. □

Thus we are left with proving the following theorem, which is where the technical details hide.

THEOREM 4.4. Let $\mathcal{E} \in \mathbf{Vect}(X_{C,E})$ be semistable of slope 0. Then there exists an injective map

$$\mathcal{O}_{X_{C,E}} \rightarrow \mathcal{E}.$$

5. DIAMONDS AND THE v -TOPOLOGY

To prove this final reduction, we will need some preliminary definitions. I will assume familiarity with adic and perfectoid spaces.

The first result is a useful motivational theorem.

THEOREM 5.1 (Scholze, [5]). Let $X/\mathbf{Q}_p$ be a rigid analytic variety. Then perfectoid spaces over X form a basis for the proétale topology.

For example, if X is “small” in the sense that there is an étale map $X \rightarrow \mathbf{T}^n$, we can use the perfectoid torus $\tilde{\mathbf{T}}^n$ to give a proétale cover.

In fact, this is even more strongly the case: picking a proétale cover $\tilde{X}$ which is perfectoid, we have

$$X = \operatorname{Coeq}(\tilde{X} \times_X \tilde{X} \rightrightarrows \tilde{X})$$

in the category of analytic adic spaces.

Now observing this is a coequalizer of perfectoid spaces, the idea is that the diamond $X^\diamond$ should generalize the tilting construction to rigid analytic spaces over $\mathbf{Q}_p$. Indeed, once we have this presentation the assignment

$$X \mapsto X^\diamond$$

should forget the structure map to $\operatorname{Spa} \mathbf{Q}_p$ by taking such a coequalizer presentation and tilting the perfectoid spaces.

This is made precise with the following definitions.

DEFINITION 5.2. Let $\mathbf{Perf}$ be the category of all characteristic p perfectoid spaces. A diamond D is a proétale sheaf on $\mathbf{Perf}$ such that

$$D = X/R$$

where $X \in \mathbf{Perf}$ and $R \subset X \times X$ is an equivalence relation such that the projections onto each copy of X are proétale.

THEOREM 5.3 (Scholze, [4, Proposition 12.3]). The category of diamonds has all products, fiber products, and quotients by pro-étale equivalence relations.

REMARK 5.4. We're using that the absolute product of characteristic p perfectoid spaces is again perfectoid.

To $X/\operatorname{Spa} \mathbf{Q}_p$ a rigid analytic space, using the previous coequalizer presentation we'd like to write

$$X^\diamond = \operatorname{Coeq}((\tilde{X} \times_X \tilde{X})^\flat \rightrightarrows \tilde{X}^\flat).$$

This doesn't literally make sense in adic space, but in the category of diamonds it does by definition. Indeed, if one interprets $(-)^{\flat}$ on a perfectoid space to mean the proétale $h_{(-)^{\flat}}$ given by the Yoneda embedding, this can be interpreted as a coequalizer in the category of diamonds. This now exists by construction.

However, it's unclear that this construction is independent of choices. A better construction is the following.

DEFINITION 5.5. Let $X/\mathrm{Spa} \mathbf{Q}_p$ be a rigid analytic space. The presheaf $X^\diamond$ on $\mathbf{Perf}$ is given by

$$S \mapsto \{(S^\#, S^\# \rightarrow X)\}.$$

Here, $S^\#$ is a characteristic zero untilt.

REMARK 5.6. If X is perfectoid and we try this, we get the sheaf for X^b under the Yoneda embedding for $\mathbf{Perf}$. This is because $\mathbf{Perfd}_X \simeq \mathbf{Perfd}_{X^b}$.

REMARK 5.7. We have $Y_{S,E} = S \times (\mathrm{Spa} E)^\diamond$, and $X_{S,E} = S/\varphi^{\mathbf{Z}} \times (\mathrm{Spa} E)^\diamond$.

THEOREM 5.8 (Scholze, [4, Section 15]). The presheaf $X^\diamond$ is a proétale sheaf on $\mathbf{Perf}$, and furthermore is a diamond. The following hold when X/K is a smooth rigid analytic space over a p -adic field:

- The functor
$$X \mapsto (X^\diamond, X^\diamond \rightarrow \mathrm{Spa} K^\diamond)$$
is fully faithful.
- We can recover $|X|$ through a presentation of the diamond via a perfectoid proétale cover $\tilde{X}$ to obtain a proétale equivalence relation $R = \tilde{X} \times_X \tilde{X}$. One has $|X| = |\tilde{X}|/|R|$.
- The category $X_{\mathrm{ét}}^\diamond$ of diamonds étale over $X^\diamond$ recovers the usual site $X_{\mathrm{ét}}$.^a

^aOne defines the diamond site $X_{\mathrm{ét}}^\diamond$ by saying $f : \mathcal{G} \rightarrow \mathcal{F}$ is étale if for $Y \rightarrow \mathcal{F}$ perfectoid the pullback $\mathcal{G} \times_{\mathcal{F}} Y$ is representable by a perfectoid space étale over Y .

We will also need to make use of a related concept called the v -topology on $\mathbf{Perfd}$ (we now use all perfectoid spaces).

DEFINITION 5.9. The v -topology on $\mathbf{Perfd}$ is the Grothendieck topology generated by open covers and all surjective maps of affinoids.

Initially it seems nothing could possibly be a sheaf for the v -topology, but actually many useful things are, including all diamonds.

THEOREM 5.10 (Scholze, [4, Section 11]). Any diamond is a v -sheaf (regarded on $\mathbf{Perf}$).

REMARK 5.11. Noting the similarity of the definition of a diamond and an algebraic space, this mirrors the result that algebraic spaces are automatically fpqc sheaves.

6. PROOF OF THE MAIN THEOREM

Recall that we proved that $\mathcal{E} \in \mathbf{Vect}(X_C)$ admitting a decomposition $\mathcal{E} \simeq \bigoplus_{\lambda \in \mathbf{Q}} \mathcal{O}(\lambda)^{n_\lambda}$ implies by the following theorem, which we will now go ahead and prove.

THEOREM 6.1. Let $\mathcal{E} \in \mathbf{Vect}(X_{C,E})$ be semistable of slope 0. Then there exists an injective map

$$\mathcal{O}_{X_{C,E}} \rightarrow \mathcal{E}.$$

Proof. We will break this proof up into several steps:

- Show that we can replace C by an extension.
- Show that after extending C , there exists $d \geq 0$ such that

$$\mathcal{O}_{X_S}(-d) \rightarrow \mathcal{E}$$

is injective.

- Reduce ruling out $d \geq 2$ to the key lemma.
- Reduce ruling out $d = 1$ to the key lemma.
- Prove the key lemma.

Step 1. Suppose the claim is true over C'/C . In this step, this actually gives a trivialization of $\mathcal{E}_{C'}$, not just a single nonzero section. Indeed, an injection

$$0 \rightarrow \mathcal{O}_{X_{C'}} \rightarrow \mathcal{E}_{C'}$$

has quotient again semistable of slope 0 in the fixed-slope category. By induction on the rank, the quotient is $\mathcal{O}_{X_{C'}}^{n-1}$, and the vanishing $\mathrm{Ext}^1(\mathcal{O}, \mathcal{O}) = 0$ from above splits the extension. Hence $\mathcal{E}_{C'} \simeq \mathcal{O}_{X_{C'}}^n$.

Considering the v -sheaf

$$S \in \mathbf{Perfd}_C \mapsto \{\mathcal{E}_S \simeq \mathcal{O}_{X_S}^n\},$$

observe that since $\Gamma_{\mathrm{pro\acute{e}t}}(S, \underline{E}) \simeq \Gamma(X_S, \mathcal{O}_{X_S})$ this is a v -quasitorsor for $\mathrm{GL}_n(\underline{E})$. Indeed on S , any continuous map $|S| \rightarrow \mathrm{GL}_n(\underline{E})$ yields an automorphism of $\underline{E}^n(S) = \mathrm{Hom}_{\mathrm{cont}}(|S|, \underline{E}^n)$. Hence we obtain an action of $\mathrm{GL}_n(\underline{E})(S)$ on $\Gamma_{\mathrm{pro\acute{e}t}}(S, \underline{E}^n) \simeq \Gamma(X_S, \mathcal{O}_{X_S}^n)$, which gives the desired action. This is only a quasitorsor because we lack v -local triviality.

Thus over C' the displayed sheaf has a section. This implies that over C we get an actual v -torsor, as we can deduce the v -local trivialization condition by the fact that $\mathrm{Spa} C' \rightarrow \mathrm{Spa} C$ is a v -cover.

Then in Scholze-Weinstein's Berkeley lectures it was shown any such $\mathrm{GL}_n(\mathbf{E})$ -torsor is representable by a perfectoid space pro-étale over $\mathrm{Spa} C$ (since $\mathrm{GL}_n(\overline{\mathbf{E}})$ is locally profinite) [8]. This implies the torsor admits a section over C , so the claim follows.

Step 2. This is where we use that $\mathcal{O}_{X_C}(1)$ is ample! Let $\mathcal{L}$ be the sub line bundle of maximal degree. Since $\mathcal{E}$ is semistable of slope zero, the degree of $\mathcal{L}$ is ≤ 0 . Thus, if $\mathcal{L}$ simply exists, we obtain an injection $\mathcal{O}_{X_S}(-d) \rightarrow \mathcal{E}$. In particular, all we need is for $\mathcal{E}$ to admit a global section after a twist. This indeed is the case by ampleness. Once we know $\mathcal{E}$ admits a sub line bundle, we can just take the maximal degree one.

Introduction of the key lemma. If $d = 0$, we are done. We will reduce cases where $d > 0$ to the key lemma below, which gives a global section and hence the desired map $\mathcal{O}_{X_{C,E}} \rightarrow \mathcal{E}$. We use step (1) to be able to take the extension to satisfy this hypothesis.

LEMMA 6.2 (Key lemma). Let

$$0 \longrightarrow \mathcal{O}_{X_C}(-1) \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_{X_C}(1/n) \longrightarrow 0$$

be an extension of vector bundles with $n \geq 1$. Then after taking some extension of C , $\mathcal{E}$ admits a global section.

Step 3. The idea is that having $d \geq 2$ contradicts minimality of d if we assume the key lemma. Since we chose the minimal d , $\mathcal{F} = \mathcal{E}/\mathcal{O}(-d)$ is again a vector bundle. It has rank $\leq n - 1$, degree d and positive slope.

Thus, using the main theorem inductively, we'll get an injection $\mathcal{O}(-d + 2) \rightarrow \mathcal{F}$ since $\mathcal{O}(-d + 2)$ has maps to $\mathcal{O}(\lambda)$ for any $\lambda \geq 0$ (recall we made a map $\mathcal{O} \rightarrow \mathcal{O}(1)$, hence to $\mathcal{O}(n)$, and $\mathcal{O}(\lambda)$ by changing E). If $d = 1$, it's possible $\mathcal{F} = \mathcal{O}(1/(n - 1))$ and we won't get a map from $\mathcal{O}(1)$ (unless $n = 2$; in many notes one skips straight to ruling out $d \geq 1$ by assuming this).

Now we apply this injection. Pulling back

$$0 \longrightarrow \mathcal{O}(-d) \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0$$

by the morphism induces an extension

$$0 \longrightarrow \mathcal{O}(-d) \longrightarrow \mathcal{G} \longrightarrow \mathcal{O}(-d + 2) \longrightarrow 0.$$

By the key lemma, after twisting to get an extension of $\mathcal{O}(-1)$ and $\mathcal{O}(1)$ after enlarging C we obtain an injection $\mathcal{O} \rightarrow \mathcal{G}(d - 1)$, and hence an injection

$$\mathcal{O}(-d + 1) \rightarrow \mathcal{G} \rightarrow \mathcal{E}$$

contradicting minimality.

Step 4. Suppose that in step 2 we obtained $d = 1$. We then get an extension

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0$$

where $\mathcal{F}$ has rank $n - 1$, degree 1, and positive slope. Via induction we can apply the classification theorem to $\mathcal{F}$. If $\mathcal{F}$ is not semistable, its maximal-slope summand gives a subbundle $\mathcal{F}' \subset \mathcal{F}$ of degree ≥ 1 and rank $\leq n - 2$. Pulling back along $\mathcal{F}' \rightarrow \mathcal{F}$ gives

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow \mathcal{E}' \longrightarrow \mathcal{F}' \longrightarrow 0.$$

The bundle $\mathcal{E}'$ has rank at most $n - 1$ and degree $\deg(\mathcal{F}') - 1 \geq 0$, so the inductive classification of $\mathcal{E}'$ gives a summand of nonnegative slope, hence a nonzero global section. Composing $\mathcal{O} \rightarrow \mathcal{E}'$ with $\mathcal{E}' \rightarrow \mathcal{E}$ contradicts the minimality of $d = 1$.

It remains to handle the case where $\mathcal{F}$ is semistable. Then induction gives $\mathcal{F} \simeq \mathcal{O}(1/(n - 1))$, and this is exactly the case covered by the key lemma.

Step 5. It remains to prove the key lemma. We're given an extension

$$0 \longrightarrow \mathcal{O}_{X_C}(-1) \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_{X_C}(1/n) \longrightarrow 0$$

and wish to show after taking an extension of C that $\mathcal{E}$ admits a global section. To avoid introducing details about diamonds, I will only give a brief sketch of the idea here. Arguing by contradiction, suppose that $\mathcal{E}$ has no global section after any extension of C . Then the connecting map in the long exact cohomology sequence gives an injection

$$\mathrm{BC}(\mathcal{O}(1/n)) \rightarrow \mathrm{BC}(\mathcal{O}(-1)[1])$$

of Banach-Colmez spaces.

One can show that $\mathrm{BC}(\mathcal{O}(1/n))$ is a perfectoid disk $\tilde{D}_C$ and

$$\mathrm{BC}(\mathcal{O}(-1)[1]) \simeq (\mathbf{A}_{C^\#}^1)^\diamond / \underline{E}.$$

However, we can argue that after base extension to C'/C

$$\tilde{D}_C \rightarrow (\mathbf{A}_{C^\#}^1)^\diamond / \underline{E}$$

is necessarily surjective, implying the map is an isomorphism. Indeed, the image hits a non-classical point (in the target classical points are totally disconnected, but the source is connected and not a point); this means after base extension the image contains a non-empty open subset of the diamond $\mathrm{BC}(\mathcal{O}(-1)[1])$.

That is, the image of the map contains an open neighborhood of the origin of the affine line in $(\mathbf{A}_{C^\#}^1)^\diamond$ after base extension, which due to the scaling action of $E^\times$ implies surjectivity.

But this cannot be the case, as it would imply the map is an isomorphism. The target $\mathrm{BC}(\mathcal{O}(-1)[1])$ is not representable but the source is by a perfectoid disk, and representability is by definition preserved under isomorphisms of diamonds.

REMARK 6.3. Given that this decomposition holds for isomorphism classes, it's a natural question to ask what exactly the difference in the categories is conceptually.

The easy answer is that some morphisms are different: in isocrystals, $\mathrm{Hom}(V_0, V_{-1}) = 0$ but $\mathrm{Hom}(\mathcal{O}, \mathcal{O}(1))$ is nonempty (as we saw with the exact sequence!).

However, there is a more interesting answer that drops any reference to isocrystals: it turns out with a modification of the curve to an “absolute curve”, we literally get an equivalence. We can contemplate the category

$$\mathrm{Bun}_{\mathrm{FF}}(X)$$

for any v -stack X of morphisms of v -stacks $X \rightarrow \mathrm{Bun}_{\mathrm{FF}}$.

By a recent theorem of Anschütz, for $\mathrm{Spa} k^\diamond$ ($k = \overline{\mathbf{F}}_q$) we actually obtain [9, Theorem 1.1]

$$\mathrm{Isoc}_{\check{E}} \simeq \mathrm{Bun}_{\mathrm{FF}}(\mathrm{Spa} k^\diamond).$$

One should think of this as “vector bundles on the absolute curve $X_{k,E}$ ”, even though such an object doesn't literally exist. The difference between the two categories then has to do with the difference between $\mathbf{Vect}(X_{k,E}) := \mathrm{Bun}_{\mathrm{FF}}(\mathrm{Spa} k^\diamond)$ and $\mathbf{Vect}(X_{C,E})$.

□

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